



Cyclic shift scheduling with on-call duties for emergency medical services

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Abstract

In workforce scheduling for emergency medical services, it is important to ensure sufficient coverage at all time. Thus, planning has to consider unpredictable employee absences. To hedge against this type of uncertainty, on-call duties can be assigned to employees. In practice, these are often assigned ex-post based on the regular schedule. Little literature on workforce scheduling for ambulances or the planning of on-call duties exists. We present new set covering based integer programming formulations for cyclic stint-based staff scheduling with on-call duties. It is desirable for employees to work on consecutive days, called a stint, with a subsequent recovery period. On-call duties can be individually scheduled in-between two stints. Our model formulations integrate different cycle times for regular and on-call duties. A simple schedule that repeats quickly is devised for regular duties, while the on-call duty schedule rotates after each cycle to ensure fairness. The proposed models are applied to a local German emergency medical services provider. Using our stint-based model formulations, the planning complexity has been greatly reduced and reasonably large problem instances can be solved to optimality. Employee preferences, such as fairness, less work on weekends and longer recovery times, were taken into account to a high degree.

Keywords Personnel planning · Emergency medical services · Cyclical scheduling · Tactical planning

1 Introduction

For emergency medical services (EMS) the workforce is one of the most expensive resources. Efficient staff scheduling is thus an important and time-consuming part of an EMS provider's business operations. The shift schedule has to assign all employees to their respective shifts on each day of the planning cycle to ensure sufficient ambulance staffing. In addition to shift coverage and organizational aspects, good schedules consider aspects such as fairness and employee satisfaction. Dissatisfaction with the shift schedule can increase staff turnover and decrease employee productivity. For the staff of an EMS department, it is desirable to work

on consecutive days with a subsequent recovery period. For instance, in Fig. 1, a complete row is the work schedule for the respective group and each block of consecutive work shifts is a stint. A stint exemplarily comprises 2 day shifts followed by 2 night shifts. Both the construction of stints and the coordination of stints and recovery periods in a schedule are constrained by some requirements (e.g. legal, practical, preference, etc).

To maintain the service level at all times, each shift has to be staffed with employees on call in addition to regular work shifts. The problem of assigning working days and hours to employees in an integrated manner is known as the tour scheduling problem. Due to the complexity of the problem, it is often difficult to solve to optimality [2].

Shift schedules can either be separately and non-cyclically generated for each planning period, or be cyclic and repeat over a certain time period [7]. Using a symmetric cyclic schedule all employees work the same schedule lagged in time [31]. Cyclic schedules are generally well suited in cases of continuous homogeneous demand structures with few different skill levels in employee qualifications. This structure is often found in fire, police and EMS departments

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Fig. 1 Exemplary asymmetric schedule for four groups and 28 days (light: day; dark: night)

g	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
1																												
2																												
3																												
4																												

[25], since services need to be ensured on a continuous basis seven days a week, 24 hours a day. Once a cyclic schedule has been found, shift planning is greatly simplified, since the planner and the employees can adhere to the same repeating schedule. The cyclic shift scheduling problem is well known and many aspects and variants have been studied [24, 31, 38].

We propose an innovative integrated model formulation to derive a nested cyclic schedule with on-call duties. The schedule repeats after one planning cycle and consists of multiple shift sequences with recovery periods in-between days on-duty. Allowable regular shift sequences are predefined by a set of stints and the planning of rest days is implicit. In stint-based scheduling, rest days additionally can be treated explicitly as stints [20, 27]. The stint-based model is formulated as a set covering model. The integration of on-call duties in cyclic shift scheduling is computationally challenging for practical problems. While on-call duties legally have to be treated the same way as regular duties, in fact, the majority is not worked. Thus, it is not desirable to plan on-call duties during regular working sequences, as they should not be disrupted by single days off-duty. Therefore, on-call duties are nested into the long recovering periods.

Our symmetric model formulation represents a sophisticated tool to quickly derive solutions to real-world sized problems. An essential feature of our integrated model formulation is the ability to determine multiple interdependent cyclic models with different cycle times. Our contribution is fourfold:

- Analysis of stint-based cyclic model formulations
- Improved formulations for symmetric cyclic models
- Nested cyclic shift schedules with on-call duties
- Successful real-world application and implementation

We have applied our models to the case of a local German EMS provider. The main concerns were to find a simple schedule that maximizes employee satisfaction and helps reduce planning complexity. Both the regular and on-call duties should be considered by the plan. Using our model formulations, a nested cyclic schedule was derived. This schedule has replaced the manual non-cyclic plan, which was previously used by the EMS department. The remainder of our paper is structured as follows. In section 2 we discuss the existing literature on staff scheduling with a focus on cyclic models. Section 3 discusses model formulations for the symmetric and asymmetric cyclic shift scheduling problem with on-call duties. Section 4 contains practical cyclic schedules derived using our model formulations and discussion about the implementation

in practice. Additionally, different settings are tested in computational experiments in Section 4.5 to demonstrate the computational advantages of the new approach.

2 Literature review

There is a rich body of literature with regard to methodology, managerial aspects and applications of personnel scheduling (for surveys, see [20] and [34]). Our review is focused on staff scheduling problems with cyclic rosters, in the scope of EMS and on-call duties.

One of the most discussed problems in staff scheduling is the nurse rostering problem. An extensive survey is given by Burke et al. [10]. Even though there is much practical research on nurse rostering in cooperation with planners, the developed scheduling systems are rarely applied and there is little academic involvement in software products [23]. Drake [16] identifies a research application gap of the nurse rostering problem, which is partly due to the many stakeholders involved in the planning process.

2.1 Cyclic staff scheduling

Cyclic or rotating schedules generate a fixed plan for each team or employee that repeats after a number of weeks [7, 25, 31]. It is especially well suited for situations with a continuous demand structure in time and a homogeneous workforce. Baker [4] gives an early review on cyclic staff scheduling differentiating between days-off scheduling, shift scheduling and tour scheduling. Days-off scheduling assigns working days to employees, while specific shifts are scheduled in a second step. Tour scheduling integrates the two problems. The (k, m) -scheduling problem of creating a minimum cost schedule of length m , where every worker works k consecutive periods, has been treated by Bartholdi et al. [6]. Emmons [18] introduces a generalization of cyclic scheduling with constraints on days-off, weekends-off and work stretch. He minimizes the number of workers necessary for seven days per week operations. Carter et al. [13] analyze the schedules of multiple emergency departments. In a case study, they find improved schedules by applying their solution approach to derive cyclic and non-cyclic schedules. Bard et al. [5] solve the cyclical nurse-rostering problem with a branch-and-price algorithm. Their model considers the possibility to hire additional traveler nurses temporarily to reduce the number of uncovered shifts as a reaction to incremental demand changes. Rocha et al. [31]

propose a new model formulation for cyclic scheduling in the cases of homogeneous and heterogeneous workforces. They demonstrate the general applicability and solution characteristics of their models for different case studies and test instances. Falcon et al. [21] use an algebraic computational approach to design and analyze cyclic schedules. Using their approach, the set of schedules, which satisfies a given set of constraints, can be enumerated and counted. Proano et al. [28] propose a multi-stage optimization approach for generating yearly rotational schedules for residency programs. Additionally, a process to compare schedules and support the implementation is proposed. The structure of personnel rostering problems is studied by Smet et al. [32]. An analysis of the problem structure for different scheduling problems shows which problems are polynomially solvable using network flow formulations. The effects of an integrated planning for on-call duties on cyclical planning models has not yet received much attention.

2.2 Stint-based scheduling

In many staff scheduling applications, the set of allowable shift sequences is highly constrained. Thus, predefined shift sequences are often utilized to generate feasible shift schedules. These stints can consist of working shifts, days-off or a mix of both. Millar and Kiragu [27] discuss network-based models that can be used to derive cyclic and non-cyclic nurse rosters. Nodes in the network represent stints, which are characterized as a pattern of consecutive shifts or days-off with a start date and length. Chew [14] solves a practical problem for apron services and identifies only a few feasible basic patterns. These basic patterns are similar to stints, since they also represent a pattern of successive shifts and days-off. Ikegami and Niwa [22] apply a decomposition approach to a nurse rostering problem, where the shift patterns are fixed for all nurses except for one at a time. The schedule for the specific nurse is then allowed to deviate from the solution to satisfy the subproblem constraints. Brucker et al. [9] use stints for a heuristic approach to nurse rostering. Each stint is penalized according to constraint violations if it is included in the schedule. Burke et al. [11] solve a multi-objective nurse rostering problem using a heuristic that generates random legal shift patterns for each nurse, while coverage violations are minimized on a second stage. Stint-based modeling is well suited for an application to EMS workforce scheduling because of the homogeneous work demand. We are not aware of any existing application to EMS.

2.3 On-call duties

On-call or standby duties are frequently used in practice to ensure service quality in cases of unplanned absences.

Planning on-call duties has not received as much attention in the literature, even though they may often have different regulations from those that apply to regular shifts. Wermus and Pope [36] deal with shift rostering for harbour pilots. In addition to work or off days, they consider standby days. For the generation of rosters, a simple heuristic is used. Nurse scheduling with on-call duties is discussed by Campbell [12]. He points out that the use of on-call overtime leads to lower cost in their scenario. Xie and Suhl [38] deal with crew rostering in the case of a public bus company. Their multi-commodity network flow model can handle both cyclic and non-cyclic crew rostering problems. In addition to assigning drivers to work duties, they also assign drivers to standby and training duties. The number of standby duties per shift is often pre-specified, as noted by Xie and Suhl [38]. El-Rifai et al. [17] deal with scheduling on-call duties as a recourse possibility for work demand coverage during seasonal epidemics in an emergency department. They find that on-call duties are a cost-efficient mean to ensure coverage in case of short-term changes in work demand. We are not aware of any study of EMS workforce scheduling that deals with on-call duties.

While on-call duties represent one way to hedge against uncertainty, Kiermaier et al. [24] use short-term flexibility to protect against uncertainty in the work demand. They model a cyclic rostering problem, where limited weekly adjustments to the starting times of employee schedules are allowed. It turns out that the problem cannot be solved as a multi-stage stochastic program. The authors use different reductions of the model, which lead to a lower level of under-coverage. Despite these theoretical advantages, in practice, it is undesirable for employees to react to short-term modifications of the schedule.

2.4 Applications to EMS

Most planning systems, which are developed for EMS, deal with location problems. A comprehensive review is recently carried out by Aringhieri et al. [3]. They note that crew scheduling has frequently been investigated in conjunction with EMS location problems. A challenging aspect is the integration of demand forecasts into workforce schedules since, as they point out, in real-life settings the quality of health-care service is often required to be constant. Bradbeer et al. [8] consider a problem, where the number and location of EMS vehicles is given. They investigate the use of a genetic algorithm for cyclic crew rostering. Their best found solutions exhibit minor violations of working time constraints. Li and Kozan [26] propose models to determine the shift start times and successive scheduling of the ambulance crews based on the working requirements. A case study with real data indicates the necessity of adjustments for a practical implementation.

Erdoğan et al. [19] study both an ambulance location and scheduling problem for crews. The ambulance location problem maximizes coverage and is solved with a tabu search algorithm. Two models are proposed for weekly scheduling of ambulance crews. The focus of their models lies on determining the number of ambulance crews working each shift and hour to maximize coverage. Rajagopalan et al. [30] propose a two-stage model, in which the first stage optimizes the expected ambulance coverage. In the second stage the required number of crews to cover the demand is determined. No individual staff schedules for each crew are derived. Vile et al. [35] develop a tool that predicts future demands and derives the required number of crews per shift so that costs are minimized. This information is then used to create a non-cyclic schedule by allocating individual crews to the shifts. Hard constraints comprise the maximum weekly working time, the maximum weekly night hours worked and rest breaks between shifts. Overtime hours are penalized as a soft constraint violation.

2.5 Implications

Many of the practical personnel scheduling models proposed in the literature are hard to solve and require specialized heuristics. In the context of ambulance crew scheduling, most works focus on vehicle locations and staff capacities, while the explicit staff scheduling receives little attention. Few works report an actual implementation of the proposed systems even though much research uses real data. Furthermore, the scheduling of on-call duties was not considered in the context of EMS to the best of our knowledge. As employees typically prefer to work on sequential days, it is undesirable to break up work sequences by over planning and considering on-call duties ex-post. On-call duties may also be subject to different constraints than regular duties. Therefore, a new stint-based integer programming formulation, which models days-off implicitly, is proposed. Using a model reduction, we can reduce the symmetric cyclic problem to a generalization of the exact covering problem. The basic formulation is extended to integrate on-call duties as part of a nested cyclic schedule. It has been successfully used to optimally solve a real-world problem and the solution is in use at a local EMS provider.

3 Mathematical model for shift scheduling

3.1 Problem description

In shift work with 24/7 operations and continuous demand, cyclic schedules are favorable due to the reduced planning effort. A cyclic schedule repeats after a predefined period of time and has to meet the staffing requirements per

shift in accordance to regulatory constraints and employee preferences. Regulatory constraints, which mainly deal with contractual working time and allowed work stretches, must be adhered to and thus are hard constraints. Employee preferences with regard to shift sequences and the number of free weekends should be taken into account where possible (see [29]). Each day can be divided into multiple predefined shift types. We consider two 12 hour shifts on a given day, however our model can easily be applied to cases with a different daily shift schedule.

Since work sequences are subject to different organizational regulations, our cyclic scheduling model is highly constrained. By using explicit binary variables to denote the employees who work each shift, a large number of constraints is necessary to model allowed shift sequences. We noticed that only a small set of shift sequences is actually feasible from a practical perspective. Therefore, we introduce a set of shift stints, which contains all shift sequences that are acceptable in practice and thus feasible for the model. We define a stint as an allowable sequence of working shifts. The proposed models represent set covering formulations in the sense that a subset has to be selected from the set of stints that covers each shift with regard to the coverage requirements. Figure 2 depicts an exemplary stint with 2 day shifts (1), followed by 2 night shifts (2). A schedule is derived by finding a sequence of multiple stints in combination with recovery periods in-between.

In cyclic scheduling, employees are often aggregated into shift groups of similar size and skill. It is desirable to size the shift groups so that each group contains the number of employees necessary to meet the demand with a surplus for vacation and other predictable absences. Thus, each shift can be covered by exactly one shift group. Our general model in Section 3.2 deals with the generation of a schedule with regular work duties, where the shift groups are scheduled so that each shift on each day is covered exactly once.

Unpredictable absences caused by sick workers are disregarded in the general model. Especially for smaller shift groups, adding up an expected value for unpredictable absences can lead to significant mismatches. Particularly in EMS, it is important to ensure sufficient coverage of the ambulances. Therefore, on-call duties are commonly used to hedge against short-term absences. If the shift group is understaffed, an employee on call fills in. Otherwise he receives a day off. On-call duties can be planned ex-post after creating a cyclic schedule for regular duties, or they can be derived in an integrated manner. To maintain high solution quality and low managerial effort, we

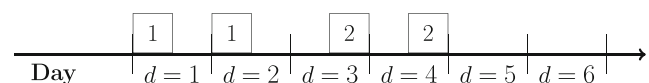


Fig. 2 Exemplary shift stint for 4 days (1 ~ day shift, 2 ~ night shift)

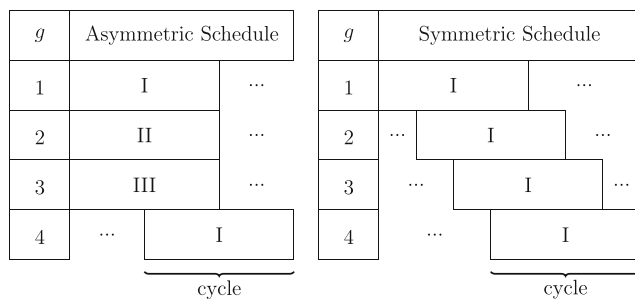


Fig. 3 Asymmetric and symmetric schedule for four groups

explicitly consider on-call duties in our integrated model in Section 3.3.

3.2 General stint-based model formulation

3.2.1 Asymmetric model

The asymmetric stint-based model *ASM* optimally schedules the shift groups $g \in G$ for a single planning cycle denoted by the cyclic set D . After one cycle D , each group works its schedule again, thus, the model is cyclic. The first element of D is considered the direct successor of the last element. The solution in the left part of Fig. 3 contains three different schedules I, II and III. Group 1 and 4 both work schedule I lagged in time, while group 2 and 3 work entirely different schedules (II and III). Thus, the roster is asymmetric. If all groups work the same schedule I shifted in time, we consider the cyclic schedule to be symmetric (see right part of Fig. 3). In this case, each group works the same shift sequences with the same number of days-off in-between. This ensures maximal equality in treatment of all staff members. In Fig. 1 and in Fig. 8 in the appendix, exemplary asymmetric and symmetric schedules with explicit shifts and identical workload are depicted.

To determine a feasible cycle length, condition (1) guarantees that the cyclic schedule repeats over the same week-days $d \in D$ for each cycle.¹² This is essential for regulatory constraints since the calendrical working hours also have to be respected for every repetition of the schedule. It is also desirable due to fairness aspects since the distribution of free weekends is the same every time the schedule repeats.

$$|D| \bmod 7 = 0 \quad (1)$$

To ensure a similar work distribution between the groups, the planning cycle D and the number of groups $|G|$ have to be chosen so that the following condition holds:

$$|D| \bmod |G| = 0 \quad (2)$$

Conditions (1) and (2) create the basis of the general model. The set of feasible sequences of day and night shifts

¹ $|D| \sim$ cardinality of D

² $x \bmod y \sim$ remainder of x after division by y

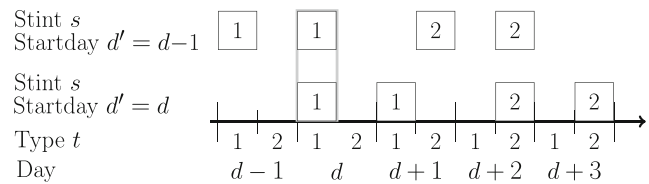


Fig. 4 Possible combinations of start day d' and stint s to cover shift $t = 1$ on day d

is defined by the stints $s \in S$. We introduce the binary variable $x_{gsd} \in \{0, 1\}$ to determine whether group g starts working stint s on day d ($x_{gsd} = 1$) or not ($x_{gsd} = 0$). As each stint defines the shift type and sequence of the included shifts, x_{gsd} needs no index for the starting shift type $t \in T$. Shift types are denoted by index set T . In our case $t = 1$ and $t = 2$ correspond to a day and a night shift respectively.

A feasible schedule for *ASM* ensures that each shift t on each day d is covered by exactly one group g . By scheduling a shift stint s on day d' all shifts contained in stint s are covered. Set Cov_{dt} contains the tuples (s, d') of stints s starting on day d' that cover shift t on day d . Figure 4 illustrates all possible combinations (s, d') of the exemplary stint s and the starting day d' of that stint to cover the day shift ($t = 1$) on day d .

Constraints (3) ensure that each shift t on each day d is covered by exactly one group g , i. e., one combination of (s, d') is selected from Cov_{dt} . By scheduling stint s on day d for a group g , the set of tuples (s, d') that can still be scheduled is reduced. No group may cover a shift that is already covered.

$$\sum_{g \in G} \sum_{(s, d') \in Cov_{dt}} x_{gsd'} = 1 \quad \forall d \in D, t \in T \quad (3)$$

Furthermore, the group that has been scheduled must not work an interleaved stint even if the shifts covered are distinct. These relations are depicted in Fig. 5. Assuming that the framed stint is part of the solution, all of the exemplary white stints (1-4) conflict with the schedule of the particular group. Stints (1) and (2) contain shifts that are distinct from the framed stint. Still, stints (1) and (2) cannot be scheduled to the particular group in conjunction with the framed stint because they are interleaved. Moreover, stints, that end directly before the start of the scheduled shift (3)

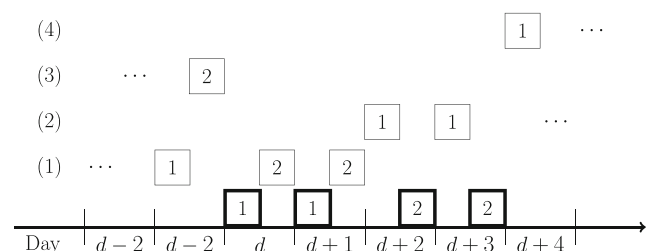


Fig. 5 Conflicting combinations of start day d' and stint s' are infeasible

and inversely (4), are not allowed, too. Thus, given that the last shift of a stint is on day $d + 3$, the earliest day to start the next stint would be $d + 5$ assuming all stints start with a day shift.

To meet these practical requirements, let E_{sd} denote the set of conflicting combinations (s, d') , when the group starts stint s on day d . Constraints (4) ensure that conflicting combinations $(s, d') \in E_{sd}$ are infeasible.

$$x_{gsd} \leq 1 - \sum_{(s', d') \in E_{sd}} x_{gs'd'} \quad \forall g \in G, s \in S, d \in D \quad (4)$$

Legal regulations and labor contracts limit the working time that a group can work per week. Let D^{Mo} be the set of days that are Mondays, and $d' \in \{d, \dots, d + 6\}$ the days of each calendrical week for $d \in D^{Mo}$. The number of scheduled shifts per calendrical week is captured by taking the sum over all tuples (s, d') that would cover a shift on any of these days. It must not exceed the maximum working time S_{cal}^{max} per calendrical week in terms of shifts (5).

$$\sum_{d' \in \{d, \dots, d+6\}} \sum_{t \in T} \sum_{(s, d'') \in Cov_{d't}} x_{gsd''} \leq S_{cal}^{max} \quad \forall d \in D^{Mo}, g \in G \quad (5)$$

Since only the working time for each calendrical week is limited (5), excessive workloads in-between weeks may occur. Thus, constraints (6) limit the number of shifts S_U^{max} worked within a given number of days U . These are applied to each sequence of length U over the planning cycle.

$$\sum_{d' \in \{d, \dots, d+U-1\}} \sum_{t \in T} \sum_{(s, d'') \in Cov_{d't}} x_{gsd''} \leq S_U^{max} \quad \forall d \in D, g \in G \quad (6)$$

In a fair schedule, each group works the same number of shifts in total and also per shift type. This ensures that no group has to work an excessive number of night shifts, which may be detrimental to the productivity of shift workers [1]. For a balanced workload, every group has to work $\frac{|D| \cdot |T|}{|G|}$ shifts in total and $\frac{|D|}{|G|}$ of each shift type t . Constraints (7) ensure an even distribution per shift type and thereby of the total workload. Parameter P_{ts} denotes the number of shifts of type t contained in each shift stint s .

$$\sum_{s \in S} \sum_{d \in D} P_{ts} x_{gsd} = \frac{|D|}{|G|} \quad \forall g \in G, t \in T \quad (7)$$

Working on weekends is one of the most important burdens in continuous shift work. Therefore, the number of free weekends, including Saturday and Sunday, constitutes an essential employee preference. Index set I denotes all weekends in the planning cycle. The variable $w_{gi} \in \{0, 1\}$ denotes whether employees in group g work ($w_{gi} = 0$) on weekend i or not ($w_{gi} = 1$). Let D_i^{we} be the set of days that define weekend i and let M be the maximum number of shifts that a shift group can work on a single weekend.

Constraints (8) force variable w_{gi} to zero if any shift is scheduled for the weekend days.

$$w_{gi} \leq 1 - \frac{1}{M} \sum_{d \in D_i^{we}} \sum_{t \in T} \sum_{(s, d') \in Cov_{dt}} x_{gsd'} \quad \forall g \in G, i \in I \quad (8)$$

The variable $\gamma \in \mathbb{N}^+$ captures the minimum number of free weekends across the groups in Eq. 9.

$$\gamma \leq \sum_{i \in I} w_{gi} \quad \forall g \in G \quad (9)$$

To ensure fairness, the minimum number of free weekends γ is maximized by the objective function (10).

$$\max \gamma \quad (10)$$

3.2.2 Symmetric model

In practice, symmetric schedules are often preferred because they ease the planning task by scheduling each group identically. Furthermore, they are inherently fair, which is desirable (see [33, 37]), since every group eventually works each part of the schedule (e.g. see Fig. 8 in the appendix).

Rocha et al. [31] describe that each group has to start the cyclic schedule with an offset of $\delta = \frac{|D|}{|G|}$ in order to guarantee the feasibility of a symmetric schedule. In fact, practical symmetric schedules satisfy this condition. In Fig. 6, an exemplary schedule with stint $1 - 1 - 2 - 2$ is illustrated for four groups with off-set $\delta = \frac{8}{4} = 2$. If we assume the offset δ is fixed, the proposed ASM can be greatly simplified to the symmetric stint-based model SSM: Constraints (7) and (9), which ensure equal distribution of work between groups and capture the minimum number of free weekends, can be dropped since the schedule for each group is identical but shifted in time. Given group $g = 1$ works any shift on day d , that particular shift is also covered on days $d + \delta(g - 1) \forall g \in \{2, \dots, |G|\}$ by the corresponding shift group g with a symmetric schedule as shown in Fig. 6. As a result, the group index g can be dropped from all variables and only one group is considered hereinafter. The schedules for the other groups are obtained by shifting

	Day									
g	1	2	3	4	5	6	7	8	1	2
1	D	D	N	N					D	D
2			D	D	N	N				
3					D	D	N	N		
4	N	N					D	D	N	N

Fig. 6 Symmetric schedule for four groups with exemplary stint

the schedule using time delay δ . In *SSM*, $x_{sd'} \in \{0, 1\}$ determines whether the group starts with stint s on day d' . Thus, the coverage constraints (3) are adjusted to (11):

$$\sum_{g \in G} \sum_{(s, d') \in Cov_{dt}} x_{sd'} = 1 \quad \forall d \in D, t \in T \quad (11)$$

Under the assumption above, the number of variables and constraints compared to the *ASM* is substantially reduced. Constraints (11) describe an exact covering problem of finding a set of stints that cover each shift exactly once.

3.3 Integrated planning of on-call duties

The general stint-based models *ASM* and *SSM* assume that the size of each shift group allows to consider predictable and planned employee absences, e.g., vacation and training, ex-post. Unplanned employee absences, e.g., due to sickness, still have to be covered in order to ensure service quality. One approach is to over-staff each shift group by the expected value of unpredictable absences plus a safety margin. This is undesirable for employees since the overstaffed workers would be off or sent home at short notice. Alternatively, the gap can be compensated by on-call duties, which only have to be worked in case of an unpredictable absence. The number of employees on call, which is required per shift, is defined by organizational rules and legal regulations. Thus, we propose an extension to our model for the integrated planning of a cyclic schedule with on-call duties. While we describe the extension of the *SSM* in the following, the *ASM* can be extended for integrated planning of on-call duties in a similar manner.

Each shift group is split into a set of subgroups SG , whereas each subgroup $h \in SG$ contains the same number of employees, a fraction of the group size, and each employee is assigned to exactly one group and subgroup. Since the subgroups of each group can be assigned to shifts independently, the flexibility of the schedule is increased. At the same time, most shifts assigned to each employee follow the quickly rotating group schedule. For an even distribution of shifts among workers, the number of days $|D|$ in the planning cycle should be chosen according to Eq. 12 in the integrated symmetric stint-based model *ISSM*. Otherwise, an allocation of on-call duties is possible but not necessarily balanced between the subgroups.

$$|D| \bmod |G| = 0 \wedge |D| \bmod |SG| = 0 \quad (12)$$

The number of on-call duties is usually relatively low compared to regular duties. Thus, they are scheduled explicitly for each day d and shift t in addition to the stint-based variables x_{sd} . The scheduling of on-call duties by subgroups h is denoted by variable $y_{hdt} \in \{0, 1\}$. Every day and night shift on every day of the planning cycle shall be staffed by OC subgroups for on-call duties as represented

by Eq. 13. The same subgroup schedule is derived for each group g with an offset of $\delta(g-1)$ days.

$$\sum_{g \in G} \sum_{h \in SG} y_{h(d+\delta(g-1))t} = OC \quad \forall d \in D, t \in T \quad (13)$$

As a result, each subgroup has to work $\frac{|D|*|T|*OC}{|SG|*|G|}$ on-call duties. If a balanced allocation of on-call duties is desirable and condition (12) holds, constraints (14) have to be respected. Otherwise they can be relaxed or disregarded.

$$\sum_{d \in D} \sum_{t \in T} y_{hdt} = \frac{|D|*|T|*OC}{|SG|*|G|} \quad \forall h \in SG \quad (14)$$

Through the provision of on-call duties, additional shifts are scheduled, while legal working time regulations do not change and the calendrical working time is still constrained by S_{cal}^{max} . In order to capture the on-call duties in the maximum number of worked shifts per week S_{cal}^{max} , constraints (5) are extended to Eq. 15:

$$\sum_{d' \in \{d, \dots, d+6\}} \sum_{t \in T} \left(\left(\sum_{(s, d'') \in Cov_{d't}} x_{sd''} \right) + y_{hd't} \right) \leq S_{cal}^{max} \quad \forall d \in D^{Mo}, h \in SG \quad (15)$$

Calendrical working time constraints do not prevent an excessive workload for each sequence of days. Thus, the general working time constraints (16) have to include on-call duties, too. In U days, a maximal number of S_U^{max} shifts, stint and on-call, is allowed.

$$\sum_{d' \in \{d, \dots, d+U-1\}} \sum_{t \in T} \left(\left(\sum_{(s, d'') \in Cov_{d't}} x_{sd''} \right) + y_{hd't} \right) \leq S_U^{max} \quad \forall d \in D, h \in SG \quad (16)$$

Furthermore, it must be ensured that for each group g and its subgroups h no on-call duties are assigned to shifts, which are part of a stint the group has to work. Additionally, on-call duties must not interleave with stints. Let $Span_{dt}$ be the set of stints, which imply that an on-call duty must not be scheduled on day d in shift t if a regular stint s starts on day d' . The following constraint prevents overlap between regular and on-call duties:

$$y_{hdt} \leq 1 - \sum_{(s, d') \in Span_{dt}} x_{sd'} \quad \forall h \in SG, d \in D, t \in T \quad (17)$$

Let S_{dt}^{con} represent the consecutive shift to shift t on day d and variable $y_{h, S_{dt}^{con}}$ a scheduled on-call duty of subgroup h in that shift. Constraints (18) are added to ensure that there cannot be two consecutive on-call duties.

$$y_{hdt} + y_{h, S_{dt}^{con}} \leq 1 \quad \forall h \in SG, d \in D, t \in T \quad (18)$$

The integration of on-call duties also affects the number of free weekends, which are maximized in accordance with employee preferences. Variable $w_{hi} \in \{0, 1\}$ now denotes

whether weekend i is free ($w_{hi} = 1$) or any shift is scheduled ($w_{hi} = 0$) for subgroup h . Let D_i^{we} be the set of days that define weekend i . The weekends off constraints (8) are modified to Eq. 19.

$$w_{hi} \leq 1 - \frac{1}{M} \sum_{d \in D_i^{we}} \sum_{t \in T} \left(\left(\sum_{(s,d') \in Cov_{dt}} x_{sd'} \right) + y_{hdt} \right) \quad \forall h \in SG, i \in I \quad (19)$$

Moreover, the integration of on-call duties may cause night to day transitions, which are not preferred by the employees. If an employee works a night shift on day d , then it is undesirable to work a day shift on day $d + 2$. Since these night to day shift transitions should be avoided, we add soft constraints regarding the four types of possible shift transitions:

- between two stints (20)
- in-between on-call duties (21)
- between on-call duty and stint shifts (22)
- between stint shifts to on-call duty (23)

Variable $r_{hd} \in \{0, 1\}$ denotes whether subgroup h works a day shift ($t = 1$) on day d that is preceded by a night shift ($t = 2$) on day $d - 2$. This allows for penalization of these undesirable shift transitions. Let $S^{day} \subseteq S$ be the set of stints that start with a day shift. Let $S^{night} \subseteq S$ be the set of stints that end with a night shift. Stints, that are not considered by S^{day} and S^{night} , are not affected by night to day transitions.

First, constraints (20) penalize night to day transitions between two consecutive stints.

$$x_{sd} + \sum_{s' \in S^{day}} x_{s'(s_{sd}^{last}+2)} \leq 1 + r_{h(s_{sd}^{last}+2)} \quad \forall s \in S^{night}, d \in D \quad (20)$$

Constraints (21) ensure that all transitions from on-call duties during night shifts to on-call duties during day shift transitions are captured by r_{hd} :

$$y_{h(d-2)2} + y_{hd1} \leq 1 + r_{hd} \quad \forall h \in SG, d \in D \quad (21)$$

Next, constraints (22) consider the transitions from an on-call duty at night time to the first shift of a stint:

$$y_{h(d-2)2} + x_{sd} \leq 1 + r_{hd} \quad \forall h \in SG, d \in D, s \in S^{day} \quad (22)$$

Finally, constraints (23) ensure that transitions from the last shift, given all possible shift blocks end with a night shift $t = 2$, in a stint to an on-call duty are included. Let s_{sd}^{last} be the last day of stint s starting on day d .

$$x_{sd} + y_{h(s_{sd}^{last}+2)1} \leq 1 + r_{h(s_{sd}^{last}+2)} \quad \forall h \in SG, d \in D, s \in S^{night} \quad (23)$$

Concerning variable r_{hd} , constraints (21)–(23) refer to the day shifts ($t = 1$) on day d . Since each subgroup

h can be scheduled for at maximum one day shift on day d , each variable r_{hd} may only allow for a single night to day transition. These undesirable transitions from night to day shifts are penalized in the objective function (24) by $\omega_2 \in \mathbb{R}^+$. Using a weighted sum, both objectives, maximizing the number of free weekends and minimizing the number of night to day rotations, are pursued.

$$\max \omega_1 \sum_{h \in SG} \sum_{i \in I} w_{hi} + \omega_2 \sum_{h \in SG} \sum_{d \in D} r_{hd} \quad (24)$$

Table 1 gives an overview of the models characterized by their names, the decision variables used, the related constraints and the sections, in which they are presented. The constraints marked by (*) are adjusted by group index g according to the discussion on asymmetric and symmetric models. The *ISSM* does not allow for a different allocation of on-call duties to subgroups across the groups. The planning of on-call duties can be integrated into the *ASM* to obtain the integrated asymmetric stint-based model (*IASM*). It allows for different subgroup schedules for each group. The asymmetric models represent a relaxation of the corresponding symmetric problem and the optimal solution for the asymmetric one may not be feasible to the symmetric model.

Comparing our stint-based formulation to one with explicitly modeled shifts, there is a trade-off between the number of variables and constraints with respect to the set of feasible stints. Using our stint-based formulation, more variables are necessary, while no constraints are required to ensure the basic shift sequence. A larger number of feasible stints implies less constraints on the shift sequences. In this case, more variables have to be introduced, while a low number of constraints can be omitted as compared to the explicit shift assignment formulation. Our model is especially suitable for a moderately sized number of feasible stints.

Table 1 Shift scheduling models with names, decision variables, related constraints and sections

	Asymmetric	Symmetric
general	<i>ASM</i> x_{gsd} (3) - (10) sect. 3.2.1	<i>SSM</i> x_{sd} (4*)-(6*), (8) (10), (11) sect. 3.2.2
integrated	<i>IASM</i> $x_{gsd}; y_{ghdt}$ (3), (4), (6), (7), (9) (10), (13*)-(24*) (sect. 3.3)	<i>ISSM</i> $x_{sd}; y_{hdt}$ (4*), (6*), (11) (13)-(24) sect. 3.3

(* ~ Constraints have to be adjusted accordingly)

4 Case study

The proposed models *SSM* and *ISSM* are applied to solve the problem of a local German EMS provider with whom a long-standing cooperation exists [15]. The monthly shift scheduling for employed paramedics has been carried out by human resources (HR) managers. An increase in operations has led to an increased ambulance demand followed by the employment of new paramedics. Thereby, the complexity of the monthly scheduling process is further increased. Using the *ISSM*, the EMS provider is able to create a fair cyclic schedule for the paramedics and to allocate on-call duties adequately.

4.1 Application scenario and practical approach

The EMS provider continuously operates 6 ambulances with 2 or 3 paramedics each, which requires 13 available paramedics per shift. To balance absences, either predictable or unpredictable, the workforce comprises 80 paramedics, who work in 2 shifts ($|T| = 2$) of 12 hours. On average, 3 paramedics are on vacation or training per shift and 4 employees are on call to compensate for unpredictable absences, which guarantees enough staff to occupy each ambulance. Thus, it is necessary to schedule 20 paramedics per shift, who are either on duty or on vacation (16 employees), and on call (4 employees). Thus, it appears natural to divide the 80 paramedics into 4 groups.

The cyclic schedule must meet the various requirements defined by the hard and soft constraints. According to employee preferences, each stint should contain at most 2 day and 2 night shifts. Furthermore, the schedule should follow a simple rhythm, which is easy to internalize (soft). According to these constraints, a practical approach was proposed by the management: A sequence of 2 day ($t=1$) and 2 night shifts ($t=2$) and 4 days-off (0) (1-1-2-2-0-0-0) could be repeated every 8 days. Each of the 4 groups starts this stint delayed by two days. The schedule meets the desired sequence and represents an extremely simple schedule. However, there must be at least one Sunday off per month to meet legal regulations (hard), which is not guaranteed by the stated schedule. This is caused by the deviation of the 8 days off-set from the 7 weekdays. Thus, the schedule actually repeats only after 7 stints and 8 weeks without accounting for the on-call scheduling ex-post. The

day of the week, on which the stint starts, rotates slowly. This leads to 5 consecutive Sundays on duty according to labor law, as a night shift on Saturday covers the morning hours of Sunday and has to be taken into account. Therefore, the approach suggested by HR managers is not feasible.

4.2 Schedule with 4 groups with selected stints

Based on the manager's approach, we applied our optimization model *SSM* to derive an optimal schedule for 4 groups. We identify two additional allowed shift stints s , which are obtained from the first soft constraint. Overall, there are only three stints, which satisfy the requirements: 1-1-2-2 ($s = 1$), 1-1-2 ($s = 2$) and 1-2-2 ($s = 3$). Dealing with 4 groups, 20 workers are allocated per shift, while 4 of them are on call. Therefore, the *SSM* is used. On-call duties have to be distributed fairly ex-post by changing the assignment of employees from regular to on-call duties.

The determined schedule comprises 28 days ($=7$ weekdays $\cdot 4$ group) according to conditions (1) and (2). The calendrical working time constraints (5) apply with a maximum of $S_{cal}^{max} = 4$ shifts per calendrical week. Furthermore, the maximum workload in shifts S_U^{max} within a given number of days U is restricted for the case study by 5 shifts within 6 days (25) and 9 shifts within 14 days (26). Though no legal regulations are violated, these workloads are considered as too excessive by management and should be avoided.

$$\sum_{d' \in \{d, \dots, d+5\}} \sum_{t \in T} \sum_{(s, d'') \in Cov_{d't}} x_{sd''} \leq 5 \quad \forall d \in D \quad (25)$$

$$\sum_{d' \in \{d, \dots, d+13\}} \sum_{t \in T} \sum_{(s, d'') \in Cov_{d't}} x_{sd''} \leq 9 \quad \forall d \in D \quad (26)$$

With these requirements, *SSM* finds no feasible solution. Therefore, an extension to the twofold planning cycle of 56 days is tested to allow for more combinations. Still, there is no feasible solution concerning the free weekends and the maximum workload. Relaxing constraints (25) and (26), the model achieves a schedule that guarantees one weekend off within 4 weeks. This schedule with 56 days is illustrated by Table 2. The filled cells signify a scheduled shift for that group, otherwise the group is off duty. The light color represent day shifts and the dark color night shifts. By scheduling this sequence for the first group and shifting it

Table 2 Schedule for 4 groups with relaxed working time constraints

d	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
1	2																											

d	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56
1	2																											

once per group g , every shift is covered. Group g begins the schedule on day $\delta \cdot (g - 1)$, with $\delta = \frac{|D|}{|G|} = \frac{56}{4} = 14$.

The schedule complies with labor law concerning the Sunday off and the calendrical working time. However, the paramedics have to work, e.g., 8 shifts within 10 days. Moreover, additional planning effort is necessary since regular and on-call duties are not differentiated and have to be assigned ex-post. To conclude, scheduling 4 groups and on-call duties by ex-post assignment cannot satisfy the stated hard and soft requirements and is thus impractical for the EMS provider.

4.3 Optimal scheduling with 5 groups and integrated on-call duties

The results of Section 4.2 show that scheduling 4 groups does not lead to satisfactory results. Therefore, we increase the flexibility of the model by separating regular duties from on-call duties using the *ISSM*. To cover the regular work demand for 16 paramedics per shift, 5 groups are build. With subgroups consisting of 2 paramedics each, we assign 2 subgroups per shift to be on call. For the planning cycle of 35 days ($= 7 \text{ weekdays} \cdot 5 \text{ groups}$), each subgroup is on call three or four times as constraint (12) does not hold. Accordingly, constraint (14) has to be relaxed. Fairness between subgroups can be achieved by rotating the scheduled on-call duties per subgroup every 35 days. The case-study specific workload constraints (25) and (26) are adopted to the *ISSM* in constraints (27) and (28) as well as the calendrical working time constraint (15) with $S_{cal}^{max} = 4$.

$$\sum_{d' \in \{d, \dots, d+5\}} \sum_{t \in T} \left(\left(\sum_{(s, d'') \in Cov_{d't}} x_{sd''} \right) + y_{hd't} \right) \leq 5 \quad \forall d \in D, h \in SG \quad (27)$$

$$\sum_{d' \in \{d, \dots, d+13\}} \sum_{t \in T} \left(\left(\sum_{(s, d'') \in Cov_{d't}} x_{sd''} \right) + y_{hd't} \right) \leq 9 \quad \forall d \in D, h \in SG \quad (28)$$

Additionally, it is not allowed to schedule 4 consecutive night shifts ($t=2$). Using only the selected stints, this practical requirement could not be violated. Under consideration of on-call duties, the requirement has to be enforced by introducing constraints (29). Thus, 2 nights of on-call duty must not succeed stint $s = 1$ or $s = 3$, which end with 2 night shifts.

$$\sum_{d' \in \{d, \dots, d+3\}} \left(\left(\sum_{(s, d) \in Cov_{d2}} x_{sd} \right) + y_{hd'2} \right) \leq 3 \quad \forall d \in D, h \in SG \quad (29)$$

Results The *ISSM* finds an optimal solution, which is depicted in Table 3. The light and dark color still denote day and night shifts, while framed cells indicate on-call duties.

The roster for 80 paramedics is obtained by shifting this schedule once per group g . The first day of the schedule is shifted for each group g by $\delta \cdot (g - 1)$ days, with $\delta = \frac{|D|}{|G|} = \frac{35}{5} = 7$. Hence, the roster covers each shift by one group with 16 paramedics and 2 subgroups.

The flexibility of scheduling subgroups independently of the main group enables the generation of a feasible schedule. Over the course of the 35 days considered by the schedule, each paramedic has at least 2 weekends off, which exceeds the requirement of the labor law and satisfies employees preferences for the whole weekend off or on duty. In addition to that, Table 3 depicts sequences of 5 days off for each subgroup respectively employee, which met great approval from the paramedics. It also improves the previous system, where HR managers tried to plan one free weekend per employee a month.

After a rhythm of 35 days, it is advisable to rotate the subgroup on call. A succession of the on-call duties can be determined manually, so that each subgroup respectively each paramedic works the same number of shifts and on-call duties. Our schedule is nested in the sense that 14 regular duties rotate over $|D|$ days and 3 to 4 on-call duties only rotate over $|SG| \cdot |D|$ days. Furthermore, the nested rotation ensures an equal working time among subgroups with regard to day and night shifts over the course of a rotation of the entire schedule.

4.4 Decision maker's point of view

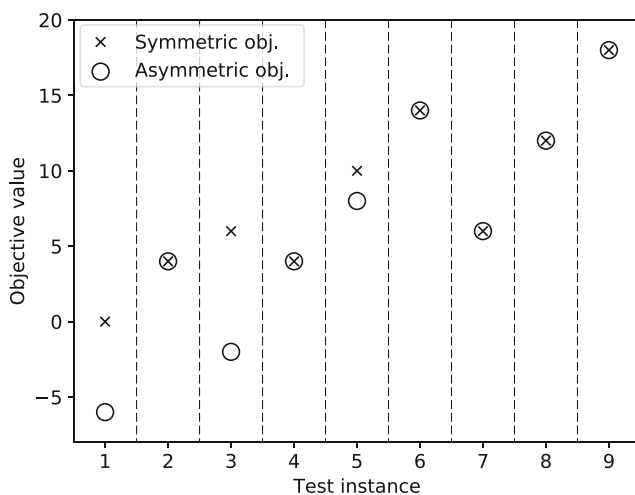
This research has been motivated by an inquiry of a local German EMS provider. An increase in workforce size has led to an increased planning complexity that overwhelmed the HR managers. The EMS department has defined three key targets: reduction of complexity in scheduling, increase of predictability for employees, and consideration of employee interests. Therefore, the decision maker suggested to introduce a symmetric cyclic schedule.

The optimal schedule obtained from *ISSM* has since replaced the previous manual planning system and it has been easily integrated into the existing information technology systems. Decision makers are convinced of this new shift scheduling framework since all addressed objectives have been met. The operational planning process has been facilitated and, thus, the managerial workload with regard to staff scheduling was reduced. In practice, there has been little need to adjust the roster ex-post and no service disruptions have occurred since implementation. Furthermore, the framework provides sufficient flexibility through on-call duties for short-term reactions, e.g., in handling of employee turnover. The paramedics appreciate

Table 4 Solution time and objective value of the asymmetric and symmetric formulation (time limit abbreviated **tl**)

No.	Parameters					<i>IASM</i>		<i>ISSM</i>	
	$ D $	$ G $	$ SG $	OC	Stints	Time (s)	Obj.	Time (s)	Obj.
1	28	4	10	1	Case	tl	−6	2.66	0
2	56	4	10	1	Case	tl	4	43.15	4
3	84	4	10	1	Case	tl	−2	511.71	6
4	35	5	8	2	Case	36.78	4	0.49	4
5	70	5	8	2	Case	tl	8	4.98	10
6	105	5	8	2	Case	505.6	14	6.07	14
7	42	6	6	3	Case	247.37	6	0.68	6
8	84	6	6	3	Case	2844.33	12	1.36	12
9	126	6	6	3	Case	2357.02	18	2.99	18
10	28	4	10	1	Full	tl	−4	17.4	1
11	56	4	10	1	Full	tl	−11	584.03	4
12	84	4	10	1	Full	tl	tl	tl	7
13	35	5	8	2	Full	1624.56	4	1.21	4
14	70	5	8	2	Full	tl	tl	12.36	10
15	105	5	8	2	Full	tl	12	13.15	14
16	42	6	6	3	Full	tl	5	3.76	6
17	84	6	6	3	Full	tl	tl	13.11	12
18	126	6	6	3	Full	tl	tl	51.96	18

ISSM solves all test instances with the selected set of stints to optimality. Optimality could not be proven in only one case with the full set of stints, while a solution was found. *IASM* could not obtain any feasible solution in 4

**Fig. 7** Best found objective (max) with asymmetric and symmetric formulation for the test instances with case stints

cases and could prove optimality in only 6 out of 18 cases. In 10 of the other 12 cases, the *ISSM* performed better. Figure 7 depicts the best found objective of both formulations within the time limit for the test instances with selected stints. This confirms the favourable computational characteristics of *ISSM*. The solution time of *ISSM* is strictly lower compared to *IASM* in all test instances.

Finally, most test instances show solution times of *ISSM*, which are sufficient for practical applications of our model. In fact, we have iteratively solved our model during a meeting with the EMS provider to adjust the schedule to their preferences.

5 Conclusion

In EMS scheduling, sufficient coverage has to be ensured at all times, therefore, on-call duties often have to be considered during shift planning. We have discussed new integer programming model formulations for practical problems with a pre-defined set of allowable shift stints and a certain demand for on-call duties. A high quality practical solution has been obtained and implemented for a local EMS provider. Employees are now guaranteed two weekends off every five weeks, as compared to one weekend a month with the previous manual non-cyclic plan. The time and cost for creating employee schedules is considerably reduced. The derived plan was seamlessly integrated into the existing scheduling software. Computational experiments indicate that our formulations are promising when solving for various scenarios with a homogeneous workforce and on-call duties in a reasonable time.

The focus of the case lies on homogeneous employees and shift demands. Employee preferences with regard to the number of free weekends and night-to-day transitions were considered. Future research should focus on the influence of additional employee preferences on model complexity. Furthermore, the implications of planning for multi-skilled employees or heterogeneous demand could be integrated into model extensions. Finally, the impact of employee vacation preferences and uncertainty aspects on the operational planning throughout the year can be evaluated.

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Appendix

Index sets

$G : g \in \{1, \dots, \mathcal{G}\}$	Set of groups
$SG : h \in \{1, \dots, \mathcal{H}\}$	Set of subgroups
$S : s \in \{1, \dots, \mathcal{S}\}$	Set of shift blocks
$S^{day} : s \subseteq S$	Set of stints that start with a day shift
$S^{night} : s \subseteq S$	Set of stints that end with a night shift
$D : d \in \{1, \dots, \mathcal{D}\}$	Set of days
$T : t \in \{1, \dots, \mathcal{T}\}$	Set of shift types
$I : i \in \{1, \dots, \mathcal{I}\}$	Set of weekends

Parameters

Cov_{dt}	Set of stints (s, d') that cover shift t on day d
E_{sd}	Set of combinations (s, d) that must not be worked if stint s starts on day d
P_{ts}	Number of shifts of type t contained in stint s
S_{cal}^{max}	Maximum number of shifts that may be worked per calendrical week
S_U^{max}	Maximum number of shifts that may be worked within a given number of days U
U	Number of days of maximum working time
S_{sd}^{last}	Last shift (d, t) of stint s starting on day d
S_{dt}^{con}	Consecutive shift (d, t) to shift t on day d
$Span_{dt}$	Set of stints that imply that no on-call duty can start on day d in shift t
D_i^{we}	Set of days d contained in weekend i
D^{Mo}	Set of days d that are Mondays
ω_1/ω_2	Weight for objective criterion 1 and 2
δ	Offset between the start of the schedule of each symmetric group
OC	On-call demand for every shift in terms of number of subgroups needed
M	Maximum number of shifts that can be scheduled on a weekend for a shift group

Decision variables

$x_{sd} \in \{0, 1\}$	1 if stint s is starting on day d , 0 else
$x_{gsd} \in \{0, 1\}$	1 if group g starts working stint s on day d , 0 else
$y_{hdt} \in \{0, 1\}$	1 if subgroup h works shift t on day d , 0 else
$y_{ghdt} \in \{0, 1\}$	1 if subgroup h of group g works shift t on day d , 0 else
$y_{h,S_{dt}^{con}} \in \{0, 1\}$	1 if subgroup h works Consecutive shift (d, t) to shift t on day d , 0 else
$w_{gi} \in \{0, 1\}$	1 if group g does not work on weekend i , 0 else
$w_{hi} \in \{0, 1\}$	1 if subgroup h does not work on weekend i , 0 else
$r_{hd} \in \{0, 1\}$	1 if subgroup h works a day shift on day d that is preceded by a night shift on day $d - 2$, 0 else
$\gamma \in \mathbb{N}^+$	Minimum number of free weekends across groups (and subgroups)

Table 5 Description of the selected and full set of stints

Stint	Selected	Full
DDDD	—	✓
DDDN	—	✓
DDNN	✓	✓
DNNN	—	✓
NNNN	—	✓
DDD	—	✓
DDN	✓	✓
DNN	✓	✓
NNN	—	✓

Fig. 8 Exemplary symmetric schedule for four groups and 28 days (light: day; dark: night; offset is highlighted)

g	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
1																												
2																												
3																												
4																												

Integrated symmetric stint-based model

$$\max \quad \omega_1 \sum_{h \in SG} \sum_{i \in I} w_{hi} + w_2 \sum_{h \in SG} \sum_{d \in D} r_{hd} \quad (24)$$

$$\text{s. t.} \quad x_{sd} \leq 1 - \sum_{(s', d') \in E_{sd}} x_{s'd'} \quad \forall s \in S, d \in D \quad (4^*)$$

$$\sum_{d' \in \{d, \dots, d+U-1\}} \sum_{t \in T} \sum_{(s, d'') \in Cov_{d't}} x_{sd''} \leq S_U^{max} \quad \forall d \in D \quad (6^*)$$

$$\sum_{g \in G} \sum_{(s, d') \in Cov_{dt}} x_{s(d'+\delta(g-1))} = 1 \quad \forall d \in D, t \in T \quad (11)$$

$$\sum_{g \in G} \sum_{h \in SG} y_{h(d+\delta(g-1))t} = OC \quad \forall d \in D, t \in T \quad (13)$$

$$\sum_{d \in D} \sum_{t \in T} y_{hdt} = \frac{|D| * |T| * OC}{|SG| * |G|} \quad \forall h \in SG \quad (14)$$

$$\sum_{d' \in \{d, \dots, d+6\}} \sum_{t \in T} \left(\left(\sum_{(s, d'') \in Cov_{d't}} x_{sd''} \right) + y_{hdt} \right) \leq S_{cal}^{max} \quad \forall d \in D^{Mo}, h \in SG \quad (15)$$

$$\sum_{d' \in \{d, \dots, d+U-1\}} \sum_{t \in T} \left(\left(\sum_{(s, d'') \in Cov_{d't}} x_{sd''} \right) + y_{hdt} \right) \leq S_U^{max} \quad \forall d \in D, h \in SG \quad (16)$$

$$y_{hdt} \leq 1 - \sum_{(s, d') \in Span_{dt}} x_{sd'} \quad \forall h \in SG, d \in D, t \in T \quad (17)$$

$$y_{hdt} + y_{h, S_{dt}^{Con}} \leq 1 \quad \forall h \in SG, d \in D, t \in T \quad (18)$$

$$w_{hi} \leq 1 - \frac{1}{M} \sum_{d \in D_i^{we}} \sum_{t \in T} \left(\left(\sum_{(s, d') \in Cov_{dt}} x_{sd'} \right) + y_{hdt} \right) \quad \forall h \in SG, i \in I \quad (19)$$

$$x_{sd} + \sum_{s' \in S^{day}} x_{s'(s_{sd}^{last}+2)} \leq 1 + r_{h(s_{sd}^{last}+2)} \quad \forall s \in S^{night}, d \in D \quad (20)$$

$$y_{h(d-2)2} + y_{hd1} \leq 1 + r_{hd} \quad \forall h \in SG, d \in D \quad (21)$$

$$y_{h(d-2)2} + x_{sd} \leq 1 + r_{hd} \quad \forall h \in SG, d \in D, s \in S^{day} \quad (22)$$

$$x_{sd} + y_{h(s_{sd}^{last}+2)1} \leq 1 + r_{h(s_{sd}^{last}+2)} \quad \forall h \in SG, d \in D, s \in S^{night} \quad (23)$$

$$x_{sd}, y_{hdt}, w_{hi}, r_{hd} \in \{0, 1\} \quad \forall d \in D, h \in SG, i \in I, s \in S, t \in T$$

$$\gamma \in \mathbb{N}$$

*Index g dropped compared to referred constraints

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